S31: Introduction to Machine Learning and Data Analysis

Course Introduction

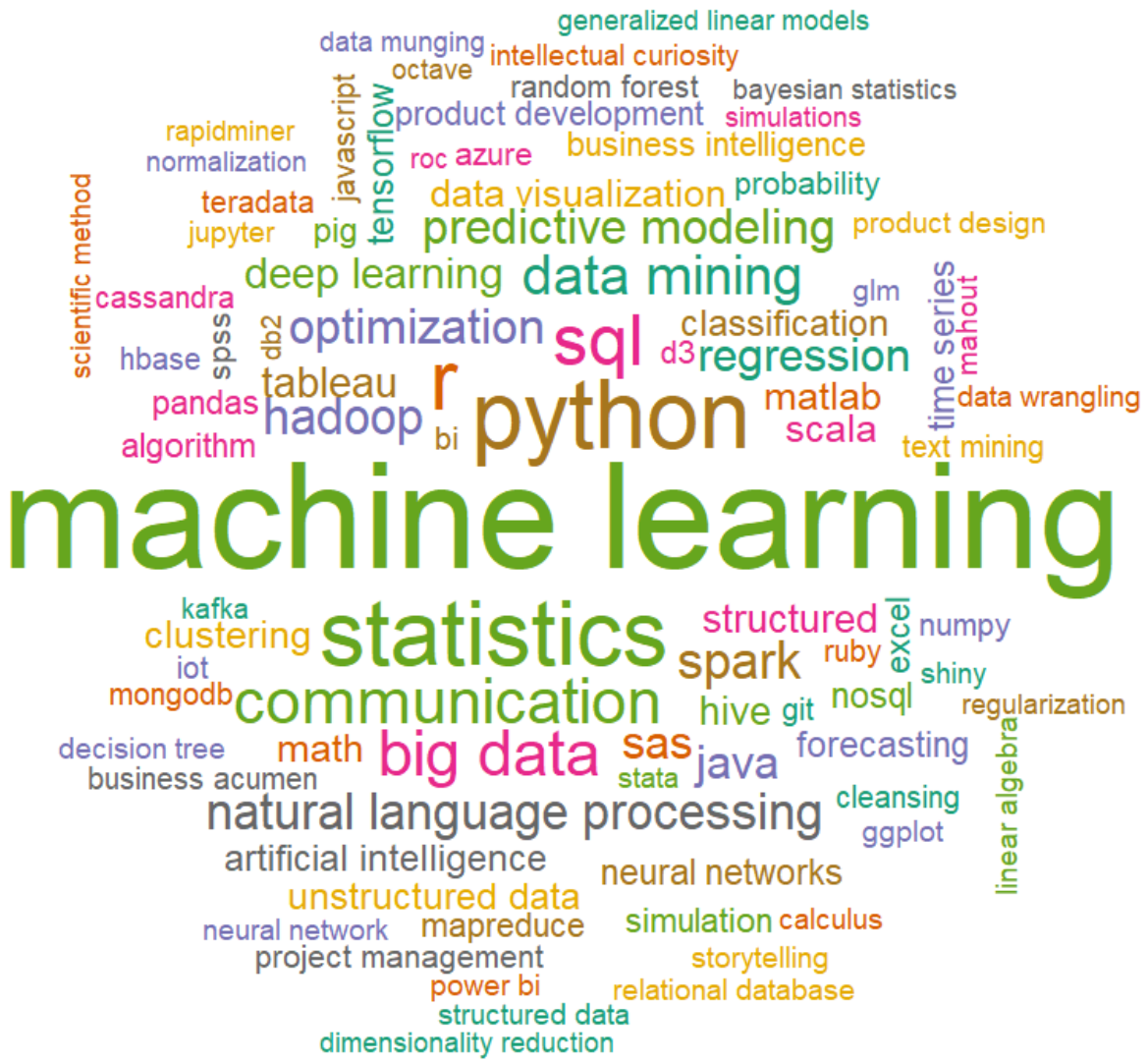
Mahdi Shafiee Kamalabad,and Maarten Cruyff

Data Science: Introduction to Machine Learning and Data Analysis

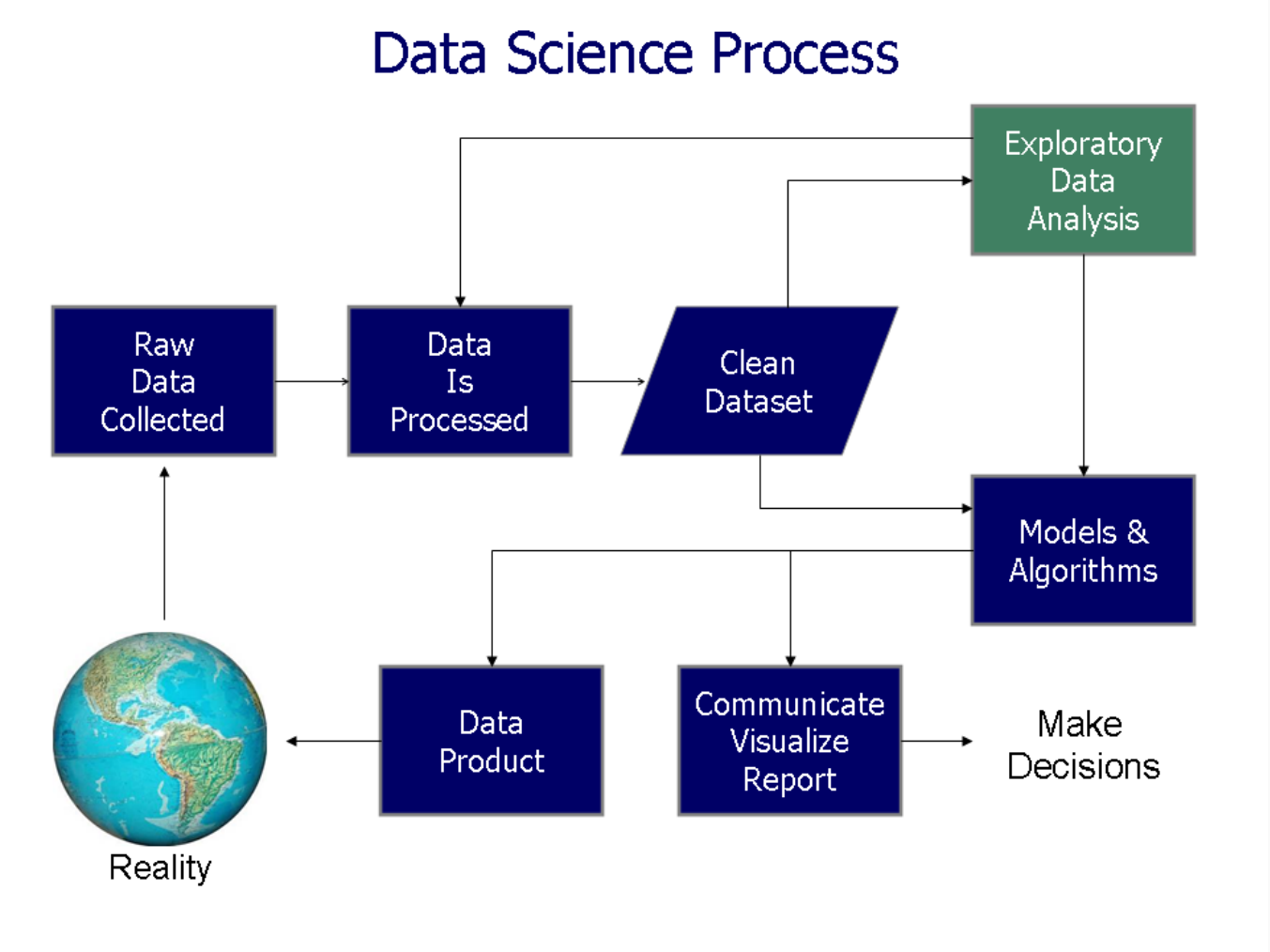
Machine Learning and Data Analysis

- **Data Analysis**
  - process of extracting insight from data
  - includes data visualization, statistical methods, and modeling
  - aims to **understand, summarize, and interpret** data
- **Machine Learning**
  - focused on developing algorithms that learn patterns from data
  - aims to **predict outcomes or support decisions** on new, unseen data
  - often emphasizes **predictive performance** more than formal hypothesis testing
- **Main types of machine learning**
  - **Supervised learning**: learn from labeled data to predict a target variable
  - **Unsupervised learning**: discover structure or patterns in unlabeled data

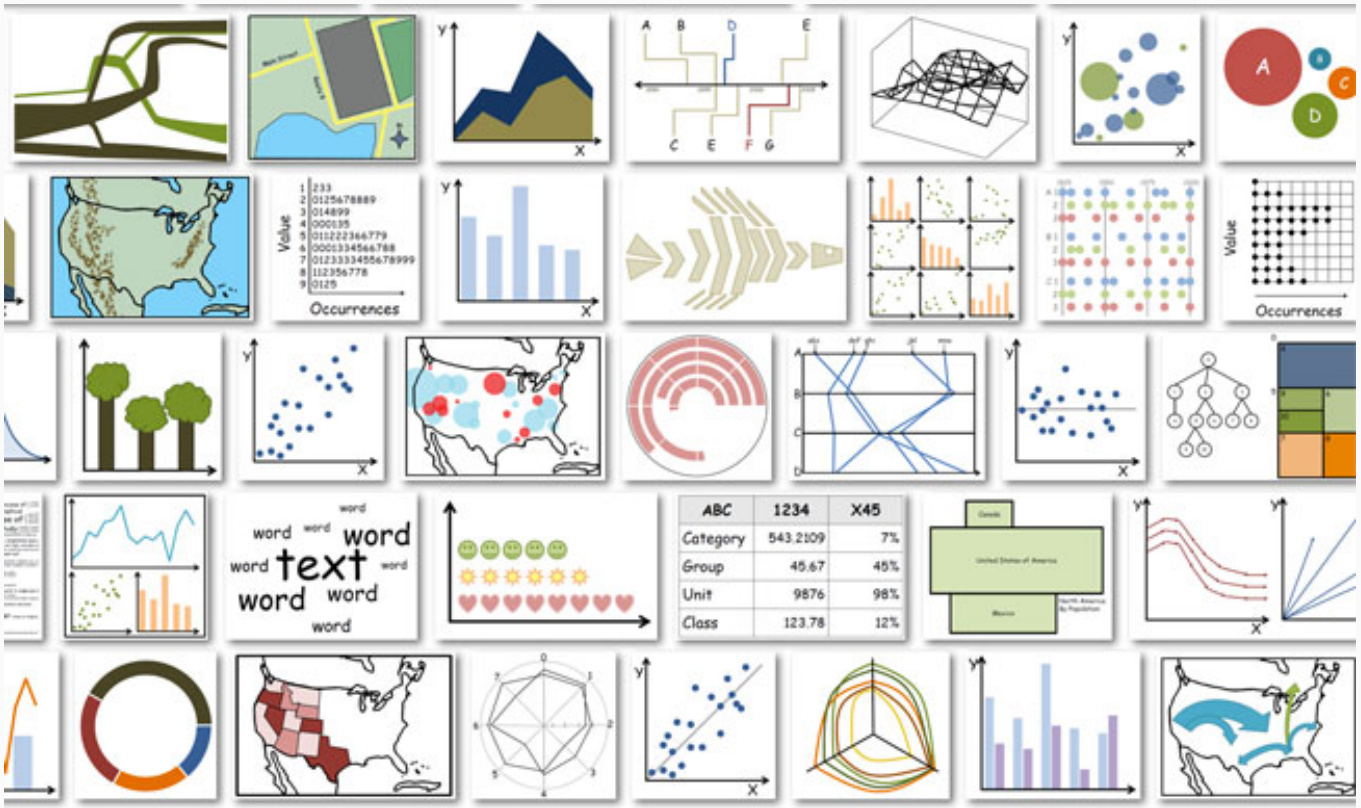
Data science word cloud



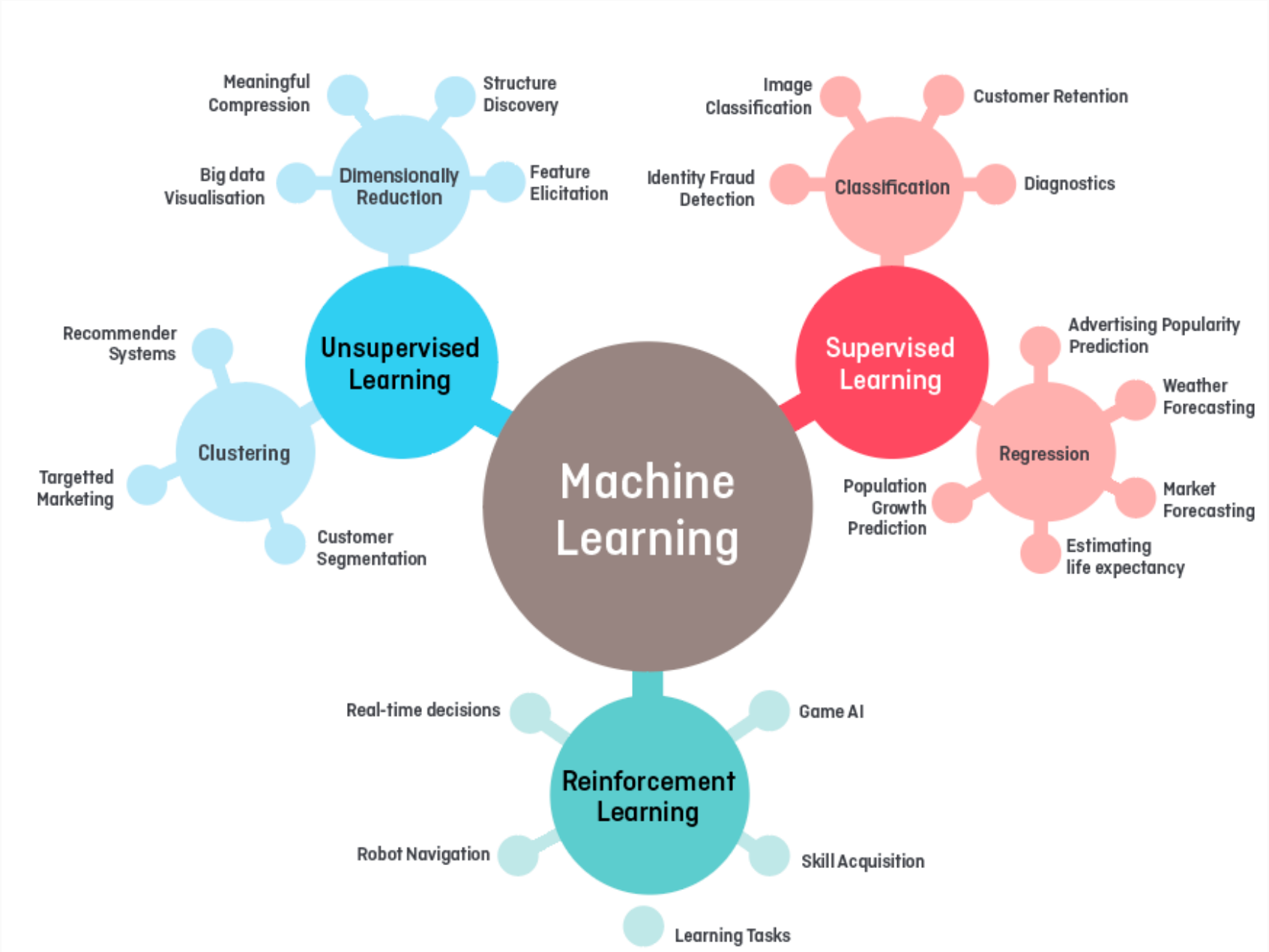
Data science process



Data visualizations



Models & Algorithms



Manual

S31: Data Analysis | Course description | Day 1 | Day 2 | Day 3 | Day 4 | Day 5

Day 1: Data science

a. Data visualization
b. Bias-variance trade-off

Day 2: Regression

a. Feature space expansions
b. Feature selection

Day 3: Classification

a. Logistic regression and LDA
b. Trees and SVMs

Day 4: Unsupervised learning

a. PCA
b. Clustering

Day 5: Deep learning

a. Neural networks
b. Presentations

Content

The course Data Analysis is part of a series of Data Science courses offered by Summerschool Utrecht. The course offers a range of statistical techniques and algorithms from statistics, machine learning and data mining to make predictions about future events and to uncover hidden structures in data. The course has a strong practical focus; participants actively learn how to apply these techniques data and to interpret the results. The course covers both classical and modern techniques of data analysis.

Course materials

Lecture slides, labs, and literature

- all available via the links in this manual

Software

You need to have installed (in that order) on your laptop:

1. R
2. RStudio

R packages

In this course we'll use a wide range of R packages. It is highly recommended that you have administrator rights for your laptop, because this will allow you to install these packages when needed. Otherwise, you should have installed these packages before the start of the course.

You can install these packages with the following code:

```
install.packages("tidyverse", "knitr", "rmarkdown", "patchwork", "palmerpenguins",
                 "patchwork", "caret", "ISLR", "splines", "glmnet", "tree",
                 "randomforest", "xgboost", "e1017", "kernlab", "pROC",
                 "psych", "tidytext", "factoextra", "neuralnet")
```

If you don't have administrator rights and you were not able to install the necessary software/packages, you can:

- work with RStudio via MyWorkPlace with your Utrecht University account
- this requires the installation of the Citrix Workspace app

Daily Schedule

Time	Activity
09.00 - 12:00	Lecture, lab, plenary discussion
12:00 - 13:00	Lunch
13:00 - 16.00	Lecture, lab, plenary discussion

Labs

Each session has a new lab. The links to the labs are:

- R lab
  - the html version of the lab
  - here you can read the exercises
  - if you get stuck, click the `SHOW` button for the R code
- R Markdown lab template
  - an R markdown document to make the exercises
  - insert R chumks for the R code
  - Knit renders an html with text, R code and output

For details on working with R Markdown see:

- R Markdown website
- R Markdown Cheat Sheet.

Recommended literature

- James, Witten, Hastie & Tibshirani. (2021). An Introduction to Statistical Learning with Applications in R, 1st ed. New York: Springer
- Wickham & Grolemund (2016). R for Data Science: Visualize, Model, Transform, Tidy and Import Data. O'Reilly
- ggplot2: Elegant Graphics for Data Analysis

Aim of the Course

By the end of this course, you should be able to:

1. Visualize data using `ggplot2`
2. Understand the basic principles of machine learning and data analysis
3. Explain the **bias-variance trade-off**
4. Recognize key models in machine learning and statistical learning
5. Understand the basic ideas behind the underlying algorithms
6. Distinguish between **supervised** and **unsupervised** learning
7. Apply these methods to real data

- The course is **introductory and less technical**
- The emphasis is on **understanding applications and practical use**

Structure of the course

- morning session (9-12)
  - lecture (45 min)
  - lab (1hr 45 min)
  - plenary discussion lab (30 min)
- lunch (12:00-13:00)
- afternoon session (12:00-16:00)
  - lecture (45 min)
  - lab (1hr 45 min)
  - plenary discussion lab (30 min)

The Labs

- The labs are presented as **HTML documents** with exercises
- The text of each exercise is visible
- The corresponding **R code** is hidden by default
- The code can be displayed by clicking the `CODE` button

The data and aesthetics arguments tell `ggplot()` where to to find the data, and where to map the variables. Together they specify the axes of the plot array, but they do not make any plot yet.

a. The first step in making plots the data specification with `ggplot(data = txhousing)`. This creates an empty plot surface.

HIDE

```
ggplot(data = txhousing)
```

b. The next step is to add one or more aesthetics. We start by mapping the variable `volume` to the x-axis. Check the result.

CODE

Remarks

- We encourage you to **first try to solve each exercise on your own**
- If you get stuck, you can consult the **code** to help you understand the solution
- You may also use **ChatGPT or similar AI tools** as additional support, but use them very critically
- If you use AI, use it to clarify concepts, debug errors, or check your understanding - not to replace your own thinking. Be critical. AI also makes mistakes...

The Rmd templates

The R labs are made with an R Markdown (`.Rmd`) document

- with empty R chunks for inserting the R code
- run code interactively by clicking the button
- render the HTML document by clicking Knit button
- use chunk options to change the default behavior of an R chunk

```
133
134 This is an R chunk with code displaying the mean of `x`
135 ```{r}
136 x <- rnorm(10, mean = 0, sd = 1)
137 mean(x)
138 ```
```

```
[1] -0.03786279
```

```
139
140 This R chunk displays a resized plot but hides the code.
141 ```{r fig.width=10, fig.heigth=8, echo=F}
142 plot(cars)
143 ```
144
```

QUESTIONS?
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